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Groin injury risk of pubertal soccer players increases during peak height velocity due to changes in movement techniques

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ABSTRACT

Adolescent athletes experience an increase in injury incidence when they undergo peak height velocity (PHV). To find the reason behind this increase, the present study investigated if PHV influences hip joint kinematics, kinetics and adductor muscle forces in two groups of adolescent soccer players performing 90°-cutting manoeuvres and inside passing. One group was estimated to be more than half a year before PHV (PRE, N = 12). The second group was estimated to be less than half a year before or after PHV (MID, N = 10). Maximum static gripping and adductor forces were measured. Motion capturing and inverse dynamics were used to calculate kinematics and kinetics. The MID group was significantly taller and heavier compared to PRE while the force measurements showed no differences. Statistics showed a higher hip abduction moment for MID during the cutting manoeuvre. Results from the anthropometrics and force measurements suggest that the moments of inertia of the participants' extremities increase faster than the muscles can adapt. A higher abduction moment of MID likely increases the load on the adductor muscles through a change of technique. Combining both findings, it is likely that the risk of suffering a groin injury is increased in the MID group.

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KEYWORDS
Sports (MeSH); biomechanical phenomena (MeSH); sprains and strains (MeSH); growth and development (MeSH); wounds and injuries (MeSH); football

1. Introduction

Children below the age of 13 participating in sports get only few injuries (Backous et al., 1988). After entering puberty, there is an increase in the injury incidence of approximately 100% (Backous et al., 1988; Volpi et al., 2003). Some studies have named a more aggressive playing style, greater risk-taking or a lack of coordination as reasons (Backous et al., 1988). While the first two reasons describe psychological factors, a lack of coordination could be seen as the result from changes in the body proportions. Around 13 years of age, the so-called peak height velocity (PHV) occurs, marking the fastest growth in body length and mass during puberty (Philippaerts et al., 2006). As a result, body proportions are changing, possibly inducing the periodically reduced coordination as mentioned above (Beunen & Malina, 1988).

Injuries in the groin area are some of the most common injuries in today's soccer (Ekstrand et al., 2011b). Groin injuries (GI) are evident in adults (Waldén et al., 2015) and children (Raya-González et al., 2019; Volpi et al., 2003), and account for 7–13% of time loss injuries (Ekstrand et al., 2011a) with a supposedly high amount of unreported cases (Harøy et al., 2017). Because GI are often recurring (Hägglund et al., 2013; Werner et al., 2009), injury prevention, especially in young athletes is important to prevent long injury histories. As is evident from the studies above, previous research has mostly concentrated on the epidemiology of GI. This has resulted in the identification of cutting manoeuvres (CM) and inside passes (IP) as movements contributing to the GI risk (Ryan et al., 2014; Whittaker et al., 2015). Nevertheless, only few biomechanical studies have investigated these movements (Chaudhari et al., 2014; Dupré et al., 2018; Franklyn-Miller et al., 2017), although understanding the biomechanical factors underlying the occurrence of GI is necessary to develop concepts for injury prevention. Regarding the connection of the aforementioned movements to GI, only little information can be found: Previous research has shown that high amounts of IP are likely to add to the risk of suffering GI by putting high stresses on the adductor muscles and their tendons (Dupré et al., 2018). On the other hand, it was shown for CM that they require a high activation of the adductor muscles (Chaudhari et al., 2014), but also require high eccentric force during the late stance phase (Serner et al., 2019), both known as factors contributing to the GI risk.

The role of maturation and its contribution to the increased injury incidence and specifically the incidence of GI has only been investigated in the recent past: A previous study of our group investigated kicking biomechanics in children regarding the risk of GI due to maturation (Dupré et al., 2020). It showed fast increasing muscle forces between different age groups around puberty performing IP in soccer. The conclusion was that the fast-changing anthropometrics and the corresponding moments of inertia require the muscle forces, and in part the kinematics, to change. It was speculated that the muscle force required to accelerate the segments increases faster than the muscles can adapt and provide the force, thereby leading to reduced angular velocities in the group nearest to PHV. Another recent study found increased leg length to be a risk factor for overuse injuries in adolescent soccer players, also attributing it to increased moments of inertia, combined with

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slow adaptations of the musculoskeletal system (Rommers et al., 2020). These findings go along with an older study which showed that soccer-playing boys were more likely to suffer an injury if they were tall but comparatively weak (Backous et al., 1988). The effects of the fast increasing moments of inertia and force requirements are likely to be the strongest during PHV. However, no previous study has investigated if PHV influences movement technique and thereby increases the injury risk.

Therefore, the purpose of this study was to investigate the hip joint kinematics, kinetics and adductor muscle forces of 90°-CM and IP in two groups of soccer-playing boys being either during or prior to their PHV. Furthermore, the purpose was to compare the aforementioned parameters, the anthropometric data and maximum adductor and gripping forces between these two groups. This will provide preliminary insights into the role of PHV, prior to longitudinal studies.

2. Materials and methods

2.1. Design

A cross-sectional design was used in this study with adolescent participants performing IP and anticipated 90°-CM. To answer the research question, the estimated time to PHV was used as an independent variable to separate the participants into two groups, based on their maturity status. Dependent variables were static force measurements, anthropometrics as well as kinematic and kinetic time series collected during IP and CM.

2.2. Participants

Twenty-two male participants were tested in the study. Their anthropometric data and maturity status can be found in Table 1. The participants were recruited from different local soccer clubs. Exclusion criteria were any chronic or acute injuries. Workload of the participants was three to four training sessions per week plus competitive matches on weekends. Because participants in this study were minors, written consent for voluntary participation was obtained by the participants themselves and by their legal guardians. The study was designed according to the requirements of the Declaration of Helsinki and was approved by the university's ethics commission (Nr. 094/2018).

2.3. Procedure

All data were collected in a motion analysis laboratory. Thirteen infrared cameras (200 Hz, 12 MX-F40 and 1 MX-3, Vicon, Oxford, GB) were used to collect kinematic data. Two force plates (1000 Hz, Kistler, Winterthur, CH), embedded in the ground, collected ground reaction forces. On top of the force plates and around them was a layer of third-generation artificial turf (Ligaturf RS Pro IICP, Polytan, Burgheim, GER) so that all movements were done on a soccer-specific surface. Before the measurement, participants were informed about the procedure by reading the subject information letter and were able to ask questions. Afterwards, anthropometric data, segment length' and maximum circumferences, were measured with a tape measure. These measurements were later also used to scale the musculoskeletal model. Sitting height and leg length were measured to calculate the time to PHV according to Mirwald et al. (2002). Participants were allowed 10 min of self-reliant warm-up.

After warm-up, two maximum force measurements were performed: Gripping force was collected with a handheld dynamometer (Jamar, Performance Health, Warrenville, IL, USA). Maximum adductor force was measured with a self-developed device, consisting of two articulating leg rests attached to a force transducer (1000 Hz) (Figure 1). The device was placed between the legs of the participants at a standardized point near the knee. Width of the device could be adjusted so that the hip angle could be kept constant at $\approx 45^{\circ}$ between different participants. For the measurement, the device was also equipped with markers to obtain the orientation of the leg rests relative to the transducer. Three measurements were performed for the gripping and the adductor measurement, each with a 1-min break between attempts.

Retro-reflective markers were then attached to anatomical reference points on the participants' skin. Afterwards, standing references were recorded from every participant. Data of 90°-CM and IP were collected in a randomized order. Every participant had to perform five valid trials of each movement.

All CM were performed to the left side, so that the transition step was always performed with the right foot. Participants were instructed to perform the CM as fast as possible. The run-out was limited in width by a line participants were not allowed to cross over to the right side. This was done to ensure that they performed a 90°-CM instead of running a curve. A trial was regarded as valid when each force plate was hit by only one foot and the participants did not step outside of the set limits of the run-out (see Figure 2(a))

IP were played as single contact passes with the dominant leg. This was determined by asking each participant which foot they used at the beginning of their soccer career. The ball

Table 1. Anthropometrics for the two groups of participants. PRE had an estimated time to PHV of >0.5 years while MID had an estimated time to PHV of < ± 0.5 years.

Group	N	Age (a)	Time to PHV (a)	Height (cm)	Mass (kg)
PRE	12	13.1 ± 0.6	−1.1 ± 0.5	156.3 ± 6.6	44.8 ± 4.7
MID	10	13.5 ± 0.4	−0.1 ± 0.3	166.4 ± 6.6	52.5 ± 5.6

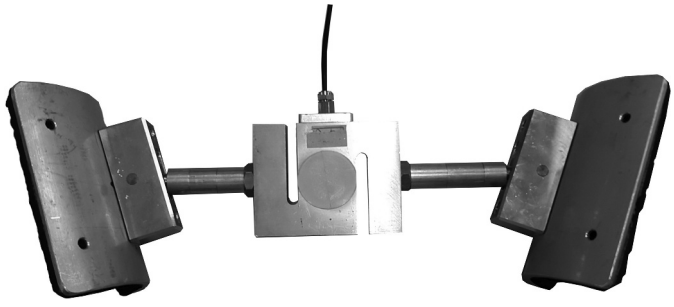

Figure 1. Image of the device used to measure the maximum isometric adductor force. It consists of a s-shaped force transducer in the middle between two articulating leg rests. Distance between the leg rests and the transducer can be adjusted in 5 mm increments.

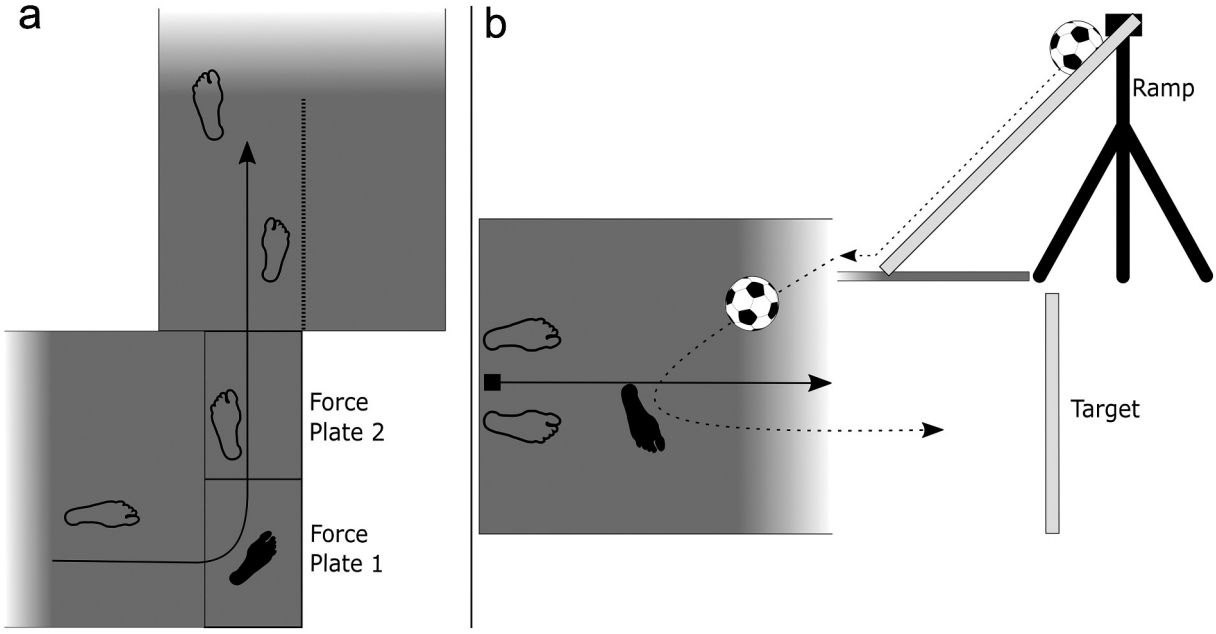

Figure 2. Schematic drawing showing test setup. The arrow depicts the direction of movement while the black foot indicates which leg was analysed for the CM (a) and IP (b). B shows the ramp that was used to accelerate the ball on the right side. The dashed arrow also indicates the movement of the ball in B.

approached the players from the front in an angle of 35° and the side from which the ball was rolled towards the participant was always opposite to their dominant leg. A ramp was used to achieve the same ball velocity of 3 ms^{-1} for every trial (Figure 2 (b)). Participants had to pass towards a target 6 m in front of them. Instructions regarding the intensity of the pass were formulated as "a pass in midfield with the friendly player being ten to fifteen metres away".

2.4. Data processing

Trials were post-processed in Vicon Nexus 2.6 (Vicon, Oxford, UK). In the IP trials, the toe marker of the shooting leg had to be virtually reconstructed as previously described (Dupré et al., 2018). AnyBody Modelling System (Version 6.0, AnyBody Technology, Aalborg, DEN) was used to perform inverse dynamics calculations on the 3D marker data. A modified version of the Anatomical Landmark Scaled Model (Lund et al., 2015) was used which has been described elsewhere (Dupré et al., 2019). Kinematic and kinetic data of CM were filtered with a recursive second-order low-pass Butterworth filter and a cut-off frequency of 20 Hz (David et al., 2017). The same filter was used for IP, but the cut-off frequency was reduced to 12.5 Hz as in previous studies on soccer biomechanics (Dupré et al., 2018). The joint moments were calculated in AnyBody Modelling System as external moments and later normalized to the participants' body mass. Segments' inertial properties and other anthropometrics are based on the work of Klein Horsman et al. (2007) and are scaled to match the participant-specific anatomy by using radial basis functions (Lund et al., 2015). Muscle forces were calculated using static optimization and a cubic cost function (Dupré et al., 2019, 2018).

Data reduction and statistics calculations were performed in Matlab 2017a (The MathWorks, Natick, Massachusetts, USA). 90°-CM were time-normalized to force plate contact. IP were time-normalized to the swing phase from toe-off of the shooting leg to ball contact. To find the maximum force value from the isometric adductor force measurements, the data were first filtered similar to the data of the movement trials. As a result of the articulated leg rests, the force of the adductors applied to the leg rests was angled in respect to the transducer whereby the transducer measured less force than applied to the leg rests. Therefore, based on the orientation obtained from the marker placement on the device, trigonometric calculations were performed to obtain the force applied at the leg rests.

2.5. Statistical analysis

To investigate not only discrete values but the continuous curve of each parameter, statistical parametric mapping (SPM) was used (Friston, 2007; Pataky et al., 2015). The analysis was performed using Matlab 2017a and the software package for SPM (Version 0.4.3, www.spm1d.org). As not all parameters were normal distributed, non-parametric permutation tests (Nichols & Holmes, 2002) were used.

For the discrete parameters from the anthropometrics and force measurements, Mann-Whitney-U-Tests were used to test for differences as some parameters showed a non-normal distribution. Furthermore, Cohen's d was calculated as a measure of the effect size for the discrete parameters. To correct for its upwards bias in small samples, a correction factor was used (Durlak, 2009). The level of significance was set to $\alpha < 0.05$ for all tests.

3. Results

The chronological age did not differ significantly between both groups (p =.054, d = .81) while body height (p = .006, d = 1.4) and mass (p = .004, d = 1.36) were significantly lower in the PRE group (Means are listed in Table 1). Estimated age at PHV was

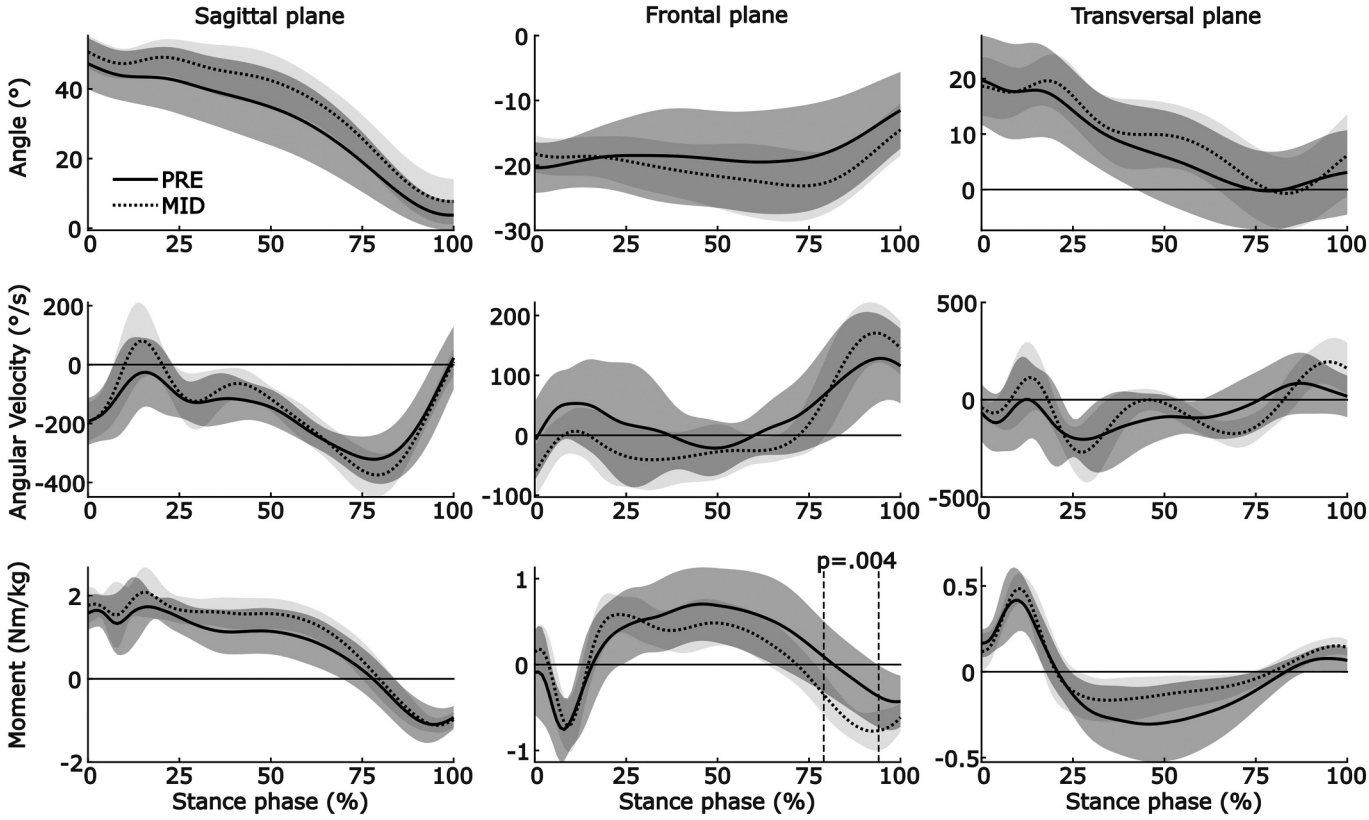

Figure 3. Mean time series during CM of PRE (N = 12, solid line) and MID (N = 10, dotted line) ± standard deviation of the hip joint angles, angular velocities and external joint moments in the three movement planes. Positive values correspond to flexion in the sagittal plane, adduction in the frontal plane and internal rotation in the transversal plane. The dashed vertical lines in the frontal plane moment indicate the interval of significant difference between PRE and MID, calculated with SPM.

statistically different with 14.2 ± 0.4 a and 13.5 ± 0.4 a for MID (p = .0.014, d = 1.11). For both groups combined, estimated age of PHV was 13.9 a. Mean static maximum adductor force was 391.77 ± 60.73 N for PRE and 442.58 ± 75.8 N for MID but not statistically different (p = .177, d = .68). Mean static maximum hand force was 269.83 ± 43.82 N for PRE and 312.1 ± 69.61 N for MID but not statistically different (p = .146, d = .67). All time series of the hip joint kinematics and kinetics of CM can be found in Figure 3. For CM, one statistical difference was found in the hip adduction moment (Figure 3), where MID showed a higher external abduction moment between 79% and 94% (p = .004) of the stance phase compared to PRE (Figure 3, second picture in bottom row). All time series of the hip joint kinematics and kinetics of IP can be found in Figure 4. The SPM analysis did not show statistical differences for any kinematic or kinetic parameter investigated in IP. Time series of adductor longus and gracilis during both movements can be found in Figure 5. There were also no statistical differences regarding the muscle forces of adductor longus and gracilis during either movement.

4. Discussion

The first aim of this study was to investigate hip joint kinematics, kinetics and adductor muscle forces during 90°-CM and IP of two groups of adolescent soccer players. To the authors' knowledge, no previous study provided comparable data of the kinematics and kinetics in all three movement planes. The second aim of this study was to investigate if hip joint kinematics and kinetics of CM or IP would differ between two groups of adolescent soccer players. One of the two groups was younger than their estimated age at PHV, while the other group was near their estimated time of PHV. Only one significant difference was found in the kinematic and kinetic time series of the hip joint in the two movements, which showed a higher external abduction moment in the MID group during the late stance phase of CM. Furthermore, it was investigated if anthropometric data as well as adductor and gripping force would be different between the two groups. The discrete anthropometric and force parameters showed a significantly greater height (+6%) and mass (+14.7%) in the MID group compared to PRE, whilst the gripping and adductor force were statistically not different.

4.1. Joint kinematics and kinetics

Due to the broad spectrum of possible cutting angles between 0 and 180° (Edwards et al., 2017; Franklyn-Miller et al., 2017), comparable research investigating the hip joint during 90°-CM is scarce. One previous study showed similar hip abduction and flexion angles for 90°-CM as the present study but did not look at further kinematics and kinetics (David et al., 2018). Furthermore, the time series from the current analysis is similar to an average of two movement strategies investigated

previously (David et al., 2018), indicating, that both strategies were employed in the MID and PRE group. 90°-CM also produces similar hip joint angles as 110°-CM (Franklyn-Miller et al., 2017). Only the transversal plane shows a different time series which might be connected to the different cutting angles: 90°-CM appears to favour more internal rotation at the hip to pre-orient the body towards the new movement direction (David et al., 2017). Because 110°-CM partly reverse the movement direction of the body, pre-orientation might be less feasible and the hip rotation is needed to re-orient the body during the stance phase. As no previous study reported the full kinematics and kinetics of the hip joint during a 90°-CM, the present data can be used as a baseline for further investigations in children.

GI have been linked to CM because of high external joint moments during mid stance requiring high activation of the adductor muscles (Chaudhari et al., 2014). Our results could not confirm this hypothesis, as mid stance showed an external hip adduction moment (Figure 3). This is plausible, as the ground reaction force vector passes the hip joint medially during mid stance of a CM to allow for a change of direction. As a result, an external adduction moment at the hip joint arises, internally countered by the abductor muscles. Hence, the two adductor muscles show none, or very low force during mid stance (Figure 5, top row). The peaks of the adductor force, and thereby peak adductor load, occurred during terminal stance of the CM. Previous research shows evidence that GI happen during terminal stance of CM, while the adductor muscles are either lengthening or are in an elongated position (Serner et al., 2019). This forces the muscles into eccentric contraction which is likely to increase the injury risk (Schache et al., 2010). Therefore, it is reasonable that the risk of suffering GI increases when performing high amounts of CM.

The kinematic time series calculated for IP were in accordance with the literature (Levanon & Dapena, 1998; Nunome et al., 2002). Muscle force time series of IP were slightly different compared to our previous studies (Dupré et al., 2018, 2020) where the adductor force was close to zero at the beginning of the swing phase. This difference can be attributed to the different test setup, were every ball came from the same, fixed position in the present study. As a result, the pass preparation and IP itself was more restricted and easier to predict compared to previous studies, which is likely to have altered the technique and the time series. Similar to our previous studies, the peak forces in both muscles occurred in the middle of the swing phase, timely close to the occurrence of the peak hip abduction angle and abduction

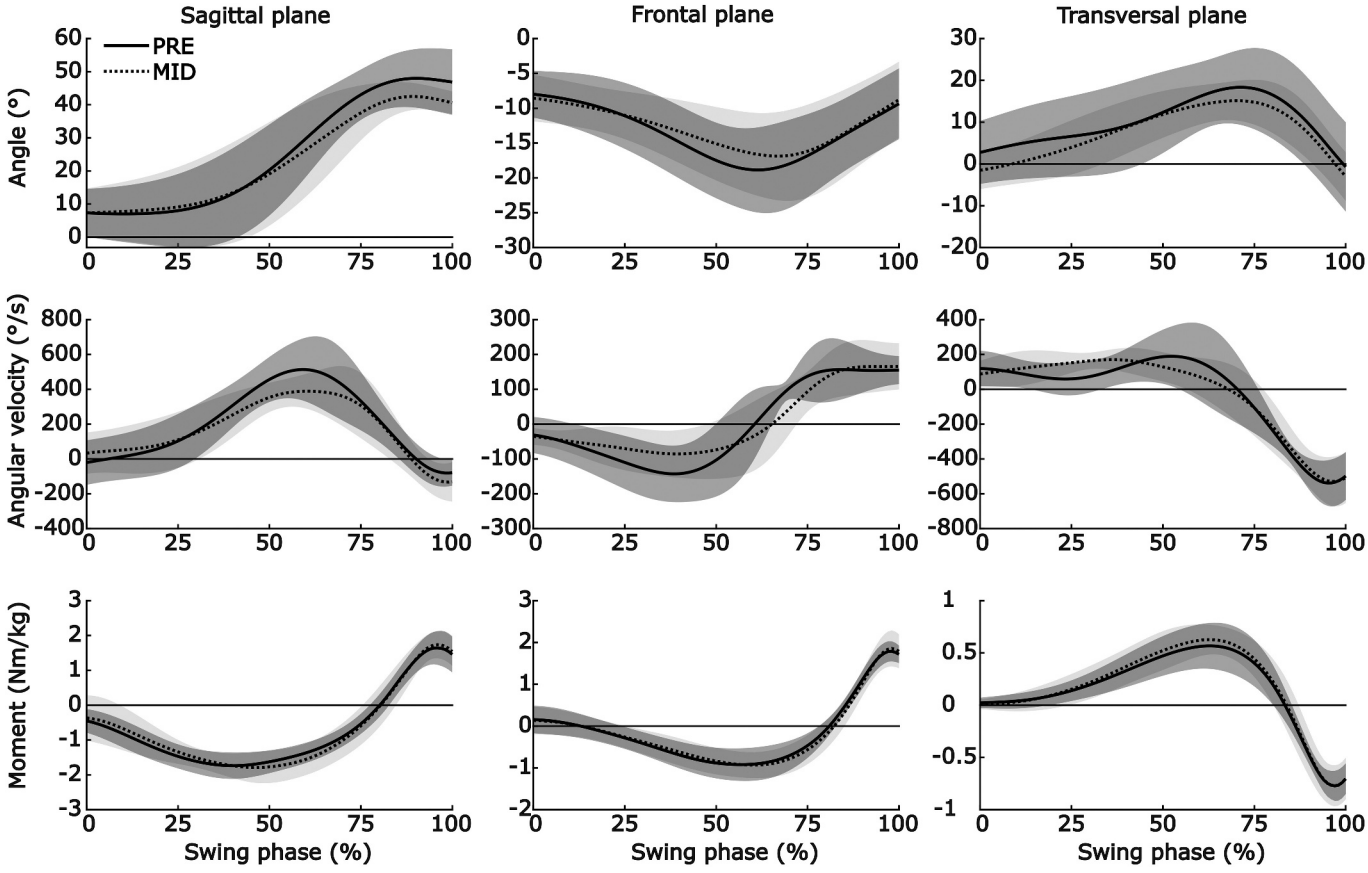

Figure 4. Mean time series during IP of PRE (N = 12, solid line) and MID (N = 10, dotted line) ± standard deviation of the hip joint angles, angular velocities and external joint moments in the three movement planes. Positive values correspond to flexion in the sagittal plane, adduction in the frontal plane and internal rotation in the transversal plane.

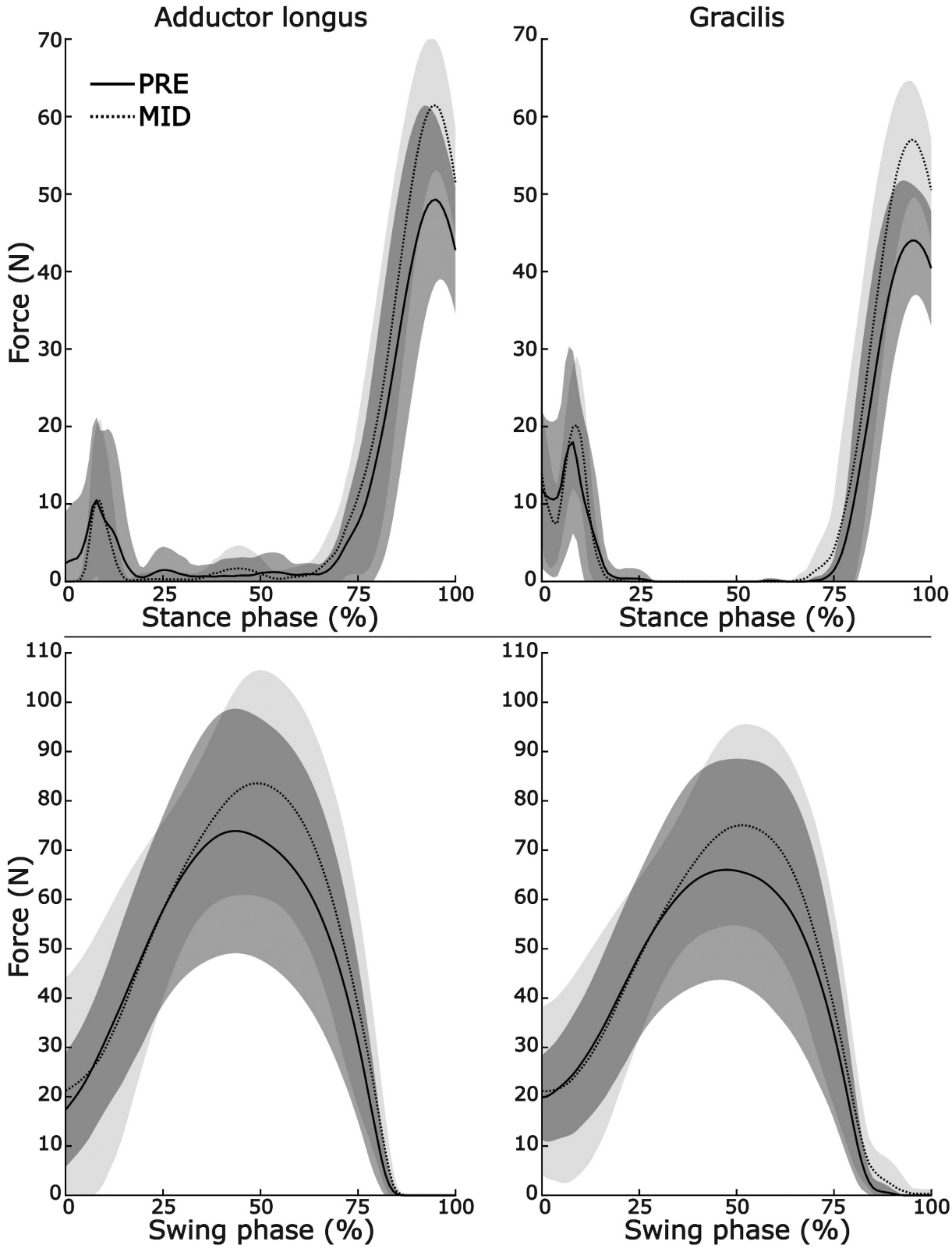

Figure 5. Mean muscle forces of PRE (N = 12, solid line) and MID (N = 10, dotted line) ± standard deviation of adductor longus and gracilis during CM (top) and IP (bottom).

velocity. In this part of the swing phase, maximum rate of stretch and maximum elongation of adductor longus occur (Charnock et al., 2009; Dupré et al., 2018). Both are known as factors that, when occurring simultaneously with high forces, increase the risk for muscle injuries (Schache et al., 2010), such as GI in this particular movement. This confirms previous assumptions that IP plays a dominant role in the development of GI (Charnock et al., 2009; Dupré et al., 2018).

4.2. Effect of growth on anthropometrics and movement technique

Mean estimated age at PHV was similar to previous studies (Bult et al., 2018; Philippaerts et al., 2006). Gripping force and height measurements have been used to investigate accelerated development during adolescence (Backous et al., 1990). Backous and colleagues associated being tall (>165 cm), due to advanced maturity, but weak (<245.25 N) with an increased injury risk (Backous et al., 1990, 1988). In our study, none of the 22 participants fit in the tall but weak category. This might be attributed to the small sample size in our study, but could also be explained by the sample investigated by Backous and colleagues, where the participants seem to have been overweight (Backous et al., 1990) and not as physically active as our population. Nevertheless, our results showed significantly increased weight and height in the MID group, with large effect sizes. The two force measurements were not significantly different but showed medium effect sizes regarding the differences between PRE and MID. This indicates an asynchronous development of the anthropometrics and muscle forces and is likely to be grounded on the same basis as the results by Backous and colleagues. In a previous study of our group, we speculated about the anthropometric parameters increasing faster than the muscular capacity and the supposed higher injury risk associated with this (Dupré et al., 2020). The moments of inertia increase synchronous with the anthropometrics, demanding higher forces from the muscles to accelerate the body's segments. If the muscles and tendons are unable to adapt quickly, overuse injuries like osteitis pubis and other GI-related injuries are likely to occur. Furthermore, such an imbalanced development might also result in changes to the movement techniques. Interestingly, the only significant difference in the time series between PRE and MID was found during terminal stance of the CM, where MID had a higher abduction moment (see Figure 3). As written above, terminal stance is the movement phase where the occurrence of GI is most likely. A higher external abduction moment of the MID group during terminal stance indicates a higher load for the adductor muscles of the respective group. From this follows that the MID group uses a movement strategy that makes greater use of the hip adductor muscles in the movement phase most common for GI. Thereby they increase their injury risk in addition to the injury risk caused by the growth spurt (Backous et al., 1988; Bult et al., 2018; Dupré et al., 2020). Contrary to the expectations, there was no significant difference in the muscle forces of CM, although MID had a higher abduction moment and significantly increased anthropometrics. MID might have reduced their acceleration during push of phase compared to PRE in order to reduce the required adductor forces.

One of our previous studies found a reduced angular velocity in IP for soccer players who were near their PHV (Dupré et al., 2020). Figure 4 shows reduced peaks in the angular velocities of MID in the sagittal and frontal plane of IP. Although the angular velocities were not significantly different in the present study, these descriptive differences in the time series resemble the previous results closely. If this could be confirmed in future studies, it would strengthen the hypothesis that the moments of inertia increase faster during PHV than the muscles can increase their force capacity, which results in reduced angular velocities. The muscle forces did not show significant differences in IP or CM with MID showing slightly higher peak forces in both muscles during both movements. As the kinematic and kinetic parameters were very similar, the increased anthropometrics of the MID group require higher muscle force due to increased inertia of the segments, making this a likely explanation for the muscle forces.

It is presumed in the literature that PHV leads to changes in movement techniques or reduced coordination (Philippaerts et al., 2006), resulting in higher injury incidences. Finding only one significant difference in the technique of CM contradicts the idea of large systematic changes in movement techniques. On the one hand, maturation has been shown to slightly influence the movement technique of IP (Dupré et al., 2020), for which only indications could be found in the present study. The fast increase of the segments' inertia was shown to cause slight reductions in the angular velocities during IP (Dupré et al., 2020), which is an open-chain movement. In CM, on the other hand, as a closed chain movement, PHV induces higher abduction moments, probably caused by slightly higher adduction velocities during terminal stance. Therefore, it cannot be ruled out that closed chain and open chain movements are affected differently during PHV. The fast-growing segments of the lower extremity and their inertia could be of more importance during open chain movements like IP as they are the only body parts that are in need of high acceleration prior to ball kicking. This is supported by a study that showed thigh and leg mass to increase disproportionately fast during adolescence (Jensen, 1989). Furthermore, it has been shown that the joint moments of open chain movements are highly affected by interaction moments due to acceleration and deceleration of distal segments (Nunome et al., 2006). An asynchronous increase of lower limb masses during adolescence would thereby add to the increased demands on the musculoskeletal system in open chain movements. In a closed chain movement like CM on the other hand, the ground reaction force is the main factor determining muscle forces and joint moments. It is, therefore, less affected by thigh mass, changing inertia and interaction moments, while the total body mass plays a more dominant role.

4.3. Limitations

The present study has some limitations: As the sample size was small in both groups, minor effects of the PHV on the technique might not have surfaced in statistics. Nevertheless, PRE and MID showed differences in the hip adduction moment and also indicated changes in joint velocities that have been shown in previous studies. This provides valuable insight despite the small sample size. Future studies with bigger samples need to refute or confirm these findings. Furthermore, longitudinal study designs are better suited to investigate effects of time and maturation than the cross-sectional approach used here. However, as no studies have investigated if PHV influences movement technique and thereby causes injuries, our preliminary, cross-sectional study provides a valuable first insight warranting large-scale future studies.

5. Conclusion

The current study shows that IP and CM both have characteristics in their motions that are likely to increase the GI risk. Furthermore, MID executes a movement technique during CM that increases the hip abduction moment in final stance, compared to PRE. This indicates a higher load on the adductor muscles. The combination of final stance, where the GI risk is the highest during CM, and the higher load on the adductor muscles increases the likeliness of a GI in the MID group during CM compared to PRE. Although it could have been expected that the significantly increased anthropometrics of the MID group would provoke further technique differences, no effect of PHV on the IP movement technique was found. Therefore, it remains unclear if PHV has a general effect on movement technique or if the effect is movement specific and discriminates between open and closed chain movements. Until the knowledge on this matter is further expanded, coaches and other practitioners should monitor the growth of pubertal players and treat players in and around PHV with care. Especially increases in training load should be postponed during this phase to prevent further loading of the musculoskeletal system. However, the use of strength training should be encouraged prior to the growth spurt to prepare players for increasing loads during PHV.

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Disclosure statement

The authors report no conflict of interest.

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